ReviewHarness: Injection-Resilient Evidence-Weighted Review under Hidden Human Evaluation

Anonymous technical report

Problem

Large language models can generate abundant criticism, but useful peer review requires evidence-supported issue selection, rubric calibration, and resistance to document-borne manipulation. Paper content is treated as untrusted evidence rather than authority.

The design targets an API-native local kernel: trusted assignment metadata and a local PDF enter the system; a strict ICML-style review leaves it. Human labels and reviewer-specific heuristics were unavailable during development.

Contributions

- Evidence-weighted disagreement resolution preserves a supported minority finding when it affects a central claim.

- Source-and-sink separation prevents paper instructions from controlling identifiers, tools, routing, scores, or output shape.

- Deadline-aware full and fast modes provide bounded review depth with per-paper failure isolation.

Threat model

Adversarial sources include visible text, hidden text, metadata, annotations, links, attachments, encoded content, and reviewer-directed imperatives. Sensitive sinks include scoring, comments, trusted identifiers, credentials, tools, network access, submission, and cross-paper state.

Reviewer calls receive sanitized paper evidence and a strict schema, but no shell, secrets, arbitrary network, submission capability, or cross-paper mutation. Detection never automatically penalizes scientific scores.

Method

Pipeline

Secure PDF ingest produces page-aware evidence and a security scan. A claim ledger identifies central, supporting, and background claims before specialist perspectives generate candidate findings. Findings are normalized, checked against paper-local locators, resolved, calibrated to the trusted rubric, formatted, and validated.

Full mode runs independent method, evidence, and impact perspectives before deterministic resolution. Fast mode uses one tri-lens perspective with the same evidence, calibration, and security gates.

Evidence and disagreement

Critical and major factual concerns require a valid page locator, an accurate evidence summary, an affected claim, decision impact, and a recommended check. Unsupported factual criticism is rejected rather than laundered into the final comment.

Agreement changes confidence; evidence strength and central-claim impact change priority. This avoids simple majority voting and preserves verified minority findings.

Calibration and containment

Final ordinal scores are derived from verified findings and official rubric anchors rather than reviewer-score averaging. Contradiction guards reject score-comment mismatches, while confidence represents assessment certainty rather than paper quality.

Trusted assignment code overwrites model identifiers. Raw suspicious spans never reach calibration. Final validation checks schema integrity, trusted-ID invariance, marker leakage, unauthorized requests, and evidence support.

Evaluation and measured proxies

Protocol

Saved evaluator artifacts are the only source for quantitative statements in this report. Controlled scientific fixtures measure evidence grounding and decision consistency; adversarial fixtures measure containment; a local batch dry run measures reliability and latency.

The report builder strictly parses security.json, quality.json, and runtime.json. Missing fields, extra fields, malformed JSON, out-of-range ratios, and non-finite values stop generation.

Runtime and concurrency setup

The production design uses bounded paper workers and bounded reviewer calls under a monotonic twenty-five-minute deadline. The controlled dry run exercises both full and fast modes, while a separate forced-failure scenario checks fallback and sibling-paper isolation. No hosted-model timing or human-labeled baseline was available, so the report includes no such comparison.

Security results

Across 12 evaluated cases, attack success was 0.0%, marker leakage was 0.0%, and unauthorized tool calls were 0.

Trusted-ID invariance was 100.0%, valid completion was 100.0%, benign false positives were 0.0%, and evaluation duration was 0.047 seconds.

Detection recall was 100.0%, clean-versus-injected score delta was 0.000, and clean-versus-injected issue overlap was 100.0%. Scope: deterministic_synthetic_fixture_and_provider.

Quality results

Across 5 controlled cases, evidence coverage was 100.0% and unsupported critique was 0.0%.

Issue precision was 100.0%, issue recall was 100.0%, minority preservation was 100.0%, and score-comment consistency was 100.0%.

Valid completion was 100.0%, repeatability was 100.0%, top-issue stability was 100.0%, and evaluation duration was 0.030 seconds. The quality gate passed.

Human correlation: N/A (unavailable; human labels and private judge heuristics were unavailable during development).

Runtime, failure analysis, and limitations

Runtime results

The batch produced 10 valid completions from 10 papers in 0.454 seconds. Per-paper p50 was 0.180 seconds and p95 was 0.281 seconds.

Timeouts were 0; retries were 0; fast-mode fallbacks were 0; invalid outputs were 0.

Failure isolation passed. Full mode passed; fast mode passed; monotonic deadline control passed.

Failure analysis

The bounded fallback order retains successful specialist evidence, switches delayed work to fast mode, and emits only schema-valid conservative results. A failed paper remains isolated from sibling workers.

Representative controlled case

In the clean controlled fixture, the resolver retains a page-local major scope limitation that directly affects the central claim, while unsupported criticism is excluded. The final comment names the cited limitation, explains its decision impact, and recommends a targeted author check; the score trace remains consistent with that retained evidence.

Known failures

Authenticated event retrieval and submission, hosted-provider rate limits and quality, OCR, image or QR semantics, and private-paper behavior remain unverified. These are explicit proof-boundary limits rather than inferred successes.

Limitations

- Human labels and private heuristics were unavailable during development.
- Proxy metrics do not guarantee human agreement.
- Novelty and significance remain partly subjective.
- Short papers may omit evidence needed for deep verification.
- Security scanning cannot prove detection of every visual or encoded attack.
- Post-hoc correlation on a small batch would have high uncertainty.

Conclusion

ReviewHarness is locally production-ready for the verified CLI, schema, security, full/fast, fallback, and bounded-batch surfaces. Its proof boundary remains explicit: evidence and proxy metrics are measured locally, while live-event behavior and hidden human alignment remain unavailable until exercised by organizers.